

Sylhet Engineering College, Sylhet
Department of Computer Science and Engineering
Term Test-01

Course: CSE 0541 1143 (Discrete Mathematics)

Marks: 10

- 1. (a)** Define a **proposition** and determine whether each of the following is a proposition. Give reasons. [03]

i) $x + 5 = 10$

ii) Dhaka is the capital of Bangladesh.

- (b)** Construct the truth table for the compound proposition: [02]

$$(p \rightarrow q) \wedge \neg q$$

- 2.** Let $A = \{1, 2, 3, 4\}$ and a relation R on A by:

$$R = \{(a, b) \in A \times A \mid a \text{ divides } b\}.$$

- (a)** Write the relation R explicitly as a set of ordered pairs. [02]

- (b)** Determine whether R is **reflexive**, **symmetric**, **antisymmetric**, and **transitive**. Provide a brief justification for each property. [03]

— End of Question Paper —